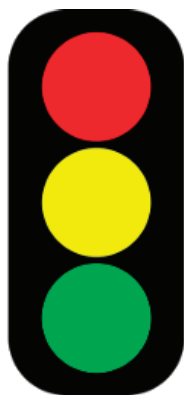




13.5.4 Wrap-Up: Reflection

1. Describe your learning today by writing a short statement for each.



Stop, I'm confused about ...

Proceed with caution,
I need clarification on ...

Go ahead. I understand ...

Unit 13, Lesson 6: Exploring Transformations – Vertical Shifts



Benchmark

MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .



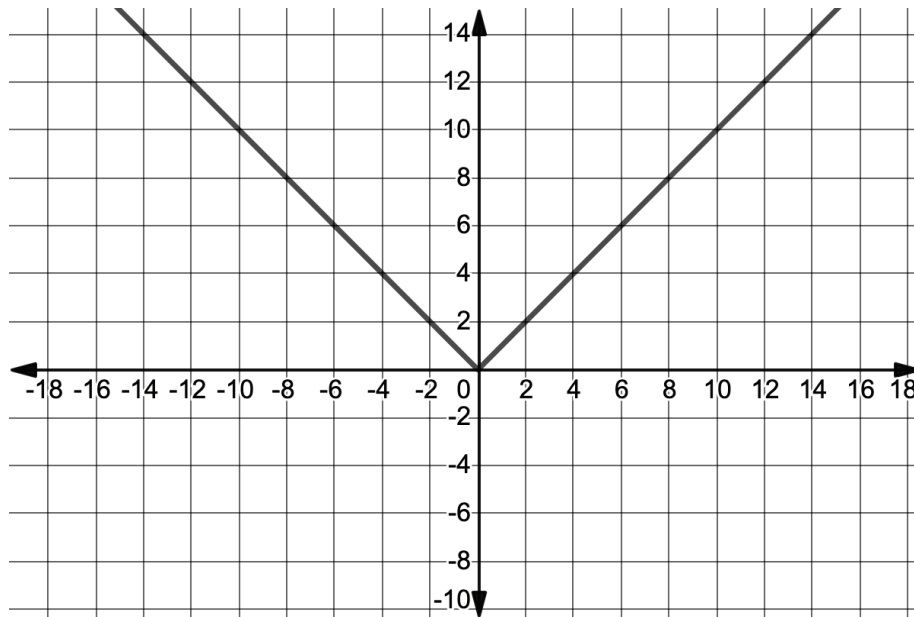
Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$.
- ☐ I can discover the effect on a table of values of $f(x)$ for a transformation of $f(x + k)$ or $f(x) + k$.



13.6.1 Warm-Up: Bell Work

1. The graph of the function, $G(x)$, is shown on the coordinate grid.



Draw or describe the graph of $G(x + 4)$.

2. The table of values of the function, $y = H(x)$, is shown.

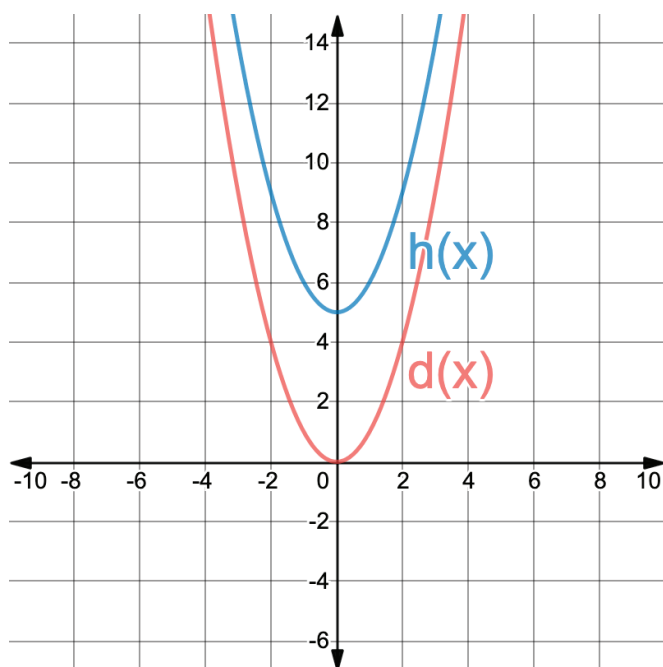
x	-8	-6	-4	0	2	6	10
H	6	3	0	-6	-3	-9	-15

Describe the effect on x -values for the function $y = H(x - 5)$.



13.6.2 Exploration: Vertical Transformations

The graphs of $y = d(x)$ and $y = h(x)$ are shown.



1. When a graph was shifted left or right, the x -values were affected. Discuss with your partner which values are affected by shifting the graph of $y = d(x)$ up 5 units to create $y = h(x)$. Summarize your discussion.
2. On the coordinate plane, draw the graph of the function $k(x)$ by shifting $d(x)$ down 3 units.

3. The table of values for $y = d(x)$, $y = h(x)$, and $y = k(x)$ are shown.

$y = d(x)$		$y = h(x)$		$y = k(x)$	
x	y	x	y	x	y
-2	4	-2	9	-2	1
0	0	0	5	0	-3
2	4	2	9	2	1

Discuss with your partner how the shift in the graph is shown in the y -values of the tables.

4. Complete the statements.

- a. The graph of $y = h(x)$ is a

☐ horizontal shift
☐ vertical shift

 of the graph of $y = d(x)$.

- b. For the graph of $y = d(x)$ to coincide with the graph

of $y = h(x)$, it is shifted

☐ down
☐ to the left
☐ to the right
☐ up

 5 units.

- c. The shift can be seen in the table of values by

☐ adding
☐ subtracting

5 to/from each

☐ x -value
☐ y -value

of

the function $d(x)$ for each corresponding

☐ x -value
☐ y -value

shown in the table.

- d. The graph of $y = k(x)$ is a

☐ horizontal shift
☐ vertical shift

of the graph of $y = d(x)$.

- e. For the graph of $y = d(x)$ to coincide with the graph

of $y = k(x)$, it is shifted

☐ down
☐ to the left
☐ to the right
☐ up

3 units.

- f. The shift can be seen in the table of values by

☐ adding
☐ subtracting

3 to/from each

☐ x -value
☐ y -value

of

the function $d(x)$ for each corresponding

☐ x -value
☐ y -value

shown in the table.

5. Write the function for each shift in terms of $d(x)$.

Up 5 units	$h(x) =$
Down 3 units	$k(x) =$

6. Ask another student how they wrote the shift in terms of $d(x)$. Ask them to justify their answer and write a summary of their justification. *You are the only person who should write in the middle column of the table.* Have the person initial next to your summary stating that your summary is accurate.

Shift	My Summary of Their Justification	Initials
Up 5 units		
Down 3 units		

7. Review the information from other students with your partner and if necessary, change the functions for each shift.



13.6.3 Bring it Together: Vertical and Horizontal Transformations

- Describe the effect on the graph and table of values of $f(x)$ for each.

Function Notation	k -values	Graph	Table of Values
$f(x + k)$	$k < 0$		
$f(x + k)$	$k > 0$		
$f(x) + k$	$k < 0$		
$f(x) + k$	$k > 0$		



13.6.4 Guided Practice: Describing Transformations

- Complete the statement to describe how the graph of $g(x) = f(x) + 6$ compares to the graph of $f(x) = |x + 8|$.

The graph of $g(x)$ represents a
☐ horizontal
☐ vertical
 shift of the graph of $f(x)$. The graph of $g(x)$ is shifted _____

units
☐ down from
☐ up from
☐ to the right of
☐ to the left of
 the graph of $f(x)$.

- Describe the effect on the x -values of $H(x) = |x + 2|$ for the transformation $K(x) = H(x - 7)$.

- Describe the effect on the y -values of $w(x) = \frac{2}{5}x - 9$ for the transformation $m(x) = w(x) + 5$.

- Describe how the graph of $P(x) = R(x + 11)$ compares to the graph of $R(x) = x^2 + 5$.



13.6.5 Wrap-Up: Growth Mindset

1. Complete the table.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always



Unit 13, Lesson 7: Exploring Transformations by a Factor



Benchmark

MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x) + k$, $kf(x)$, $f(kx)$, and $f(x + k)$ for specific values of k .



Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(kx)$ or $kf(x)$.
- ☐ I can discover the effect on the table of values of $f(x)$ for a transformation of $f(kx)$ or $kf(x)$.